Contents

1 Day 1. Classic binomial and multinomial identities

Planned entrance quiz :)

1. How many ways are to order 6 persons in a queue?

2. The box contains 50 balls numbered from 1 to 50. We draw 6 balls without replacement and write down the resulting sequence of numbers. How many different sequences are possible?

3. I have 5 yellow books and 7 red books. I want to place them on my bookshelf and I would like to place yellow books to the left of red books. How many book orders are possible?

Definition 1.1. Factorial of n, denoted by $n!$, is the number of ways to put n distinct objects in a queue.

Definition 1.2. Binomial coefficient $\binom{n}{k}$ is the number of ways to select k distinct objects from n distinct objects.

Definition 1.3. For $n = a + b + c$ the multinomial coefficient $\binom{n}{a,b,c}$ is the number of ways to split n distinct objects into different boxes, where the first box contains a objects, the second box contains b objects, the third box contains c objects.

One may create multinomial coefficient for any number of boxes :)

Exercise 1.1. Is $\binom{n}{k}$ equal to $\binom{n}{n-k}$?

Is $\binom{n}{k}$ equal to $\binom{n}{k,n-k}$?

Exercise 1.2. Write the product

$$\binom{100}{20} \cdot \binom{80}{50} \cdot \binom{30}{10}$$

as one multinomial coefficient.

Exercise 1.3. Write

$$\binom{n}{a, b, c, d}$$

using binomial coefficients.

Write

$$\binom{n}{a, b, c, d}$$

using factorials of n, a, b, c and d.

Idea 1. Solution strategies for combinatorial identities:

1. Double counting or story proof.

2. Visual identity (rare gem!).
3. Evaluation of a polynomial at a special point. Differentiation, integration.
4. Combine old identities into new.
5. Guess the answer and verify by induction.
6. Dark art!

Pascal's triangle and rule

Exercise 1.4. Recall the Pascal arithmetic triangle. Draw two versions: triangular and rectangular.

Exercise 1.5. Pascal's rule. Simplify

$$\binom{n}{k} + \binom{n}{k+1}.$$

$$\binom{n}{a-1, b, c} + \binom{n}{a, b-1, c} + \binom{n}{a, b, c-1}.$$

$$\binom{n}{a-1, b, c, d} + \binom{n}{a, b-1, c, d} + \binom{n}{a, b, c-1, d} + \binom{n}{a, b, c, d-1}.$$

Exercise 1.6. Simplify

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Recall the Pascal's rule and simplify

$$\binom{2n}{0} + \binom{2n}{2} + \binom{2n}{4} + \cdots + \binom{2n}{2n}.$$

Simplify

$$\binom{2n-1}{0} + \binom{2n-1}{2} + \binom{2n-1}{4} + \cdots + \binom{2n-1}{2n-2}.$$

Exercise 1.7. Simplify

$$\sum_{a+b+c=2026} \binom{2026}{a, b, c}.$$

$$\sum_{a+b+c+d=2027} \binom{2027}{a, b, c, d}.$$

$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

$$\sum_{a+b\leq n} \binom{n}{a} \cdot \binom{n-a}{b}.$$

Here a, b and c are assumed non-negative integers.

Hockey-stick identity

Exercise 1.8. Hockey-stick identity, Christmas stocking identity, boomerang identity.
Simplify

$$\sum_{n=a}^{b} \binom{n}{a}.$$

$$1 + 2 + 3 + \cdots + n.$$

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n \cdot (n+1).$$

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \cdots + n \cdot (n+1) \cdot (n+2).$$

Exercise 1.9. Obtain closed form expressions for:

$$1^2 + 2^2 + \cdots + n^2,$$

$$1^3 + 2^3 + \cdots + n^3,$$

$$1^4 + 2^4 + \cdots + n^4.$$

Exercise 1.10. Hockey-sticks plus idea from linear algebra!

a) Decompose $n^2 + 2n + 5$ using $\binom{n}{0}$, $\binom{n}{1}$, $\binom{n}{2}$ as the basis.

b) Find the sum

$$\sum_{k=2}^{2026} k^2 + 2k + 5.$$

c) Using similar trick find the sums

$$\sum_{k=1}^{n} 5k^2 + 4k, \quad \sum_{k=1}^{n} k^3 + 2k^2.$$

Exercise 1.11. Let's start from some number $\binom{n}{k}$ inside the Pascal's triangle.
In one step we can go either in north-east or north-west direction.
Find the sum of all numbers we can reach.

Binomial or Taylor or MacLaurin expansion

Exercise 1.12. Write the Taylor expansion (binomial expansion) at $x = 0$ for

$$(1+x)^n, \quad (x+y)^n.$$

Exercise 1.13. By plugging some special points to the $(x+y)^n$ or somewhere else find the sums

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

$$\sum_{k=0}^{n} k\binom{n}{k}, \quad \sum_{k=0}^{n} (k-1)\binom{n}{k}, \quad \sum_{k=0}^{n} (-1)^k \binom{n}{k}, \quad \sum_{k=0}^{n} (-1)^{k-1} k \binom{n}{k}$$

Hint: differentiation may help you.

How that can be?

Exercise 1.14. Be brave, really brave!

a) Take a deep breath!

b) Calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

Exercise 1.15. a) Simplify

$$\binom{-1/2}{3} + \binom{-1/2}{4}.$$

b) Draw the Pascal's triangle (the whole half-plane!) starting from $\binom{-1/2}{0}$.

c) Draw the Pascal's triangle (the whole half-plane!) starting from $\binom{1/3}{0}$.

Exercise 1.16. Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

Exercise 1.17. Simplify

$$\binom{10}{0}\binom{-0.5}{7} + \binom{10}{1}\binom{-0.5}{6} + \binom{10}{2}\binom{-0.5}{5} + \cdots + \binom{10}{7}\binom{-0.5}{0}.$$

Hint: for a honest proof consider the binomial expansion of $(1+x)^10$, $(1+x)^{-0.5}$, $(1+x)^{10-0.5}$.

Exercise 1.18. Homage to Newton.

a) Write the Taylor expansion at $x = 0$ for $(1-x^2)^{1/2}$.

b) Visually (or analytically if you wish) find the integral

$$\int_{-1}^{1} (1-x^2)^{1/2}.$$

c) Calculate the sums

$$\binom{1/2}{0} - \frac{1}{3}\binom{1/2}{1} + \frac{1}{5}\binom{1/2}{2} + \ldots$$

$$\sum_{k=2}^{\infty} \frac{1}{2k+1} \frac{(2k-3)!!}{(2k)!!}$$

Home assignment-1:

1. 1.7(3) in the old enumeration:

   Simplify
   $$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

2. 1.9(2) in the old enumeration:

   Find explicit formula for the sum
   $$1^3 + 2^3 + \cdots + n^3.$$
   Hint: represent the general term k^3 as the sum of binomial coefficients and then use the hockey stick or Christmas stocking or boomerang!

3. 1.10(2) in the old enumeration:

   [*] Using hockey stick decomposition find the sums
   $$\sum_{k=1}^{n} 5k^2 + 4k.$$

4. 1.19 in the old enumeration:

   [*] Take a deep breath and calculate
   $$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

5. 1.21 in the old enumeration:

   Write the Taylor expansion at $x = 0$ for
   $$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

6. Watch a Veritasium video «Newton changed the game»:

   https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA.

You are more than welcome to solve other exercises that we skipped. Starred exercises will be on the quiz next day.

2 Day 2. Wandermonde's identity and beyond

Wandermonde's identity

Exercise 2.1. Simplify
$$\binom{a}{0}\binom{b}{m} + \binom{a}{1}\binom{b}{m-1} + \binom{a}{2}\binom{b}{m-2} + \cdots + \binom{a}{m}\binom{b}{0}.$$

$$\binom{a}{0}\binom{m}{0}+\binom{a}{1}\binom{m}{1}+\binom{a}{2}\binom{m}{3}+\cdots+\binom{a}{m}\binom{m}{m}.$$

$$\binom{a}{0}^2+\binom{a}{1}^2+\binom{a}{2}^2+\cdots+\binom{a}{a}^2.$$

Exercise 2.2. There are $n = 500$ fishes in the pond, $t = 30$ of them are tagged. You capture $c = 5$ fishes at random without replacement. Let the random variable Y be the number of captured tagged fishes. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 2.3. There are $n = 24$ candies in the jar, 14 with orange flavour and 10 with ananas flavour. Your take 6 candies at random. Let Y be the number of candies with orange flavour you picked. Find the probabilities $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$, $\mathbb{P}(Y = y)$.

Exercise 2.4. A bridge hand (random 13 cards out of 52-card standard deck) is dealt.

a) What is the probability that the hand contains no spades?

b) What is the probability that the hand contains exactly 5 hearts?

c) What is the probability that the hand contains at most two aces?

Exercise 2.5. Wandermonde-like identities in a funny notation. Here I intentionally omit the running index i, I write the first summation term and specify with $\uparrow$ and $\downarrow$ which terms goes up or down in the sum.

a) Simplify and rewrite in a standard notation

$$\sum \binom{n}{0\uparrow}\binom{n\downarrow}{k}.$$

b) Simplify and rewrite in a standard notation

$$\sum \binom{a\uparrow}{a}\binom{n\downarrow}{k}.$$

c) Simplify and rewrite in a standard notation

$$\sum \binom{b}{0\uparrow}\binom{n}{k\downarrow}.$$

Exercise 2.6. Simplify

$$\sum_{a+b+c=k} \binom{n_1}{a}\binom{n_2}{b}\binom{n_3}{c}.$$

Gauss summation trick

Exercise 2.7. Simplify

$$1+2+3+\cdots+n;$$

$$1^2+2^2+3^2+\cdots+n^2;$$

$$\binom{n}{1}+2\binom{n}{2}+3\binom{n}{3}+\cdots+n\binom{n}{n};$$

$$\binom{n}{0}+2\binom{n}{1}+3\binom{n}{2}+\cdots+(n+1)\binom{n}{n};$$

$$\binom{n}{2}+2\binom{n}{3}+3\binom{n}{4}+\cdots+(n-1)\binom{n}{n};$$

$$\binom{n}{0}+3\binom{n}{1}+5\binom{n}{2}+\cdots+(2n+1)\binom{n}{n};$$

$$\binom{n}{0}-2\binom{n}{1}+3\binom{n}{2}-4\binom{n}{3}\cdots+(-1)^n(n+1)\binom{n}{n};$$

$$3\binom{n}{1}+7\binom{n}{2}+11\binom{n}{3}+\cdots+(4n-1)\binom{n}{n};$$

$$\binom{n}{1}-2\binom{n}{2}+3\binom{n}{3}-4\binom{n}{4}+\cdots+(-1)^n n\binom{n}{n};$$

Stars and bars

Exercise 2.8. There are 20 chairs in a row and 7 students. Each student needs one chair to sit down. We do care only about which chairs are occupied.

a) In how many ways chairs can be occupied?

b) In how many ways chairs can be occupied if there should be at least one empty chair between occupied chairs?

c) In how many ways chairs can be occupied if there should be at least two empty chairs between occupied chairs?

Exercise 2.9. I have four bags of different colors and 20 identical marble stones.

a) In how many different ways can I put marbles into bags?

b) In how many different ways can I put marbles into bags if there should be no empty bags?

Theorem 2.1 (Stars and bars theorem)**.** The number of ways to distribute n identical objects into k distinguishable bags is equal to

$$\binom{n+k-1}{n}=\binom{n+k-1}{k-1}.$$

Hint: n is the number of stars and $(k-1)$ is the number of bars.

Exercise 2.10. There are 5 types of post-cards sold at the post-office.

a) In how many different ways can I buy 12 post-cards?

b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?

Exercise 2.11. Consider the equation $x_1 + x_2 + x_3 + x_4 = 2026$.

a) How many solutions does it have if x_1, x_2, x_3 and x_4 are non-negative integers?

b) How many solutions does it have if x_1, x_2, x_3 and x_4 are positive integers?

c) How many solutions does it have if all x_i are integers and $x_i \geq 2$ for all i?

d) How many solutions does it have if all x_i are integers and $x_i \geq i$ for all i?

Exercise 2.12. There are 10 standard indistinguishable dices.

a) How many different combinations of scores are possible?

b) How many different combinations of scores are possible if at least one dice shows 6?

c) How many different combinations of scores are possible if no dice shows 5 nor 6?

Exercise 2.13. Stars, bars, circles and probability!
Consider a well shuffled deck of 52 cards. I open the cards one by one.

a) How many cards on average I open to see the first Queen?

b) How many cards on average I open to see the third Ace?

c) What is expected number of cards between the third and the fourth King?

d) What is the probability that the third Ace will be the 20-th card opened?

e) Write combinatorial identities hidden in a), b) and c).

Visualize it

Exercise 2.14. a) Express as a binomial coefficient and visualize the sum:

$$1 + 2 + 3 + 4 + \cdots + n.$$

b) Visualize the Fuji-sum and simplify it!

$$1 + 2 + 3 + \cdots + (n-1) + n + (n-1) + \cdots + 3 + 2 + 1.$$

c) Recall the multiplication table for numbers from 1 to 10. Why the sum of all numbers is just

$$1^3 + 2^3 + 3^3 + \cdots + 10^3?$$

Simplify the sum!

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[Har+10] John J Harris et al. Combinatorics and graph theory. Springer-Verlag New York, 2010. Accessible book with a lot of pictures, stories and nice explanations, 355 pages.

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